Feng Qian

Curriculum Vitae

(614) 270-8786
feng.qian@tufts.edu
github:fqian5

Education

Aug. 2021 –Aug. 2026 **Ph.D. in Physics**, Tufts University, Medford, Massachusetts.
Advisor: Peter J. Love

Aug. 2017 –May 2021 **B.S. in Physics with Honors Research Distinction**, Magna Cum Laude, minor in Honor Math, The Ohio State University, Columbus, Ohio.
Advisor: Daniel Gauthier

Publication

Warm Start Adiabatic State Preparation by Clock Hamiltonian(In preparation).

Universal Parent Hamiltonian for Preparable State (In preparation).

Pauli Based Diagonalization using Mielke Similarity Renormalization Group (In preparation).

Contextuality supplies the correlation energy for simulating chemical Hamiltonians (In preparation).

Quantum Correlation in the Linear Optical Circuit for Quantum Fourier Transform.
Honor Undergraduate Thesis

Presentations

- **Warm Start Adiabatic State Preparation**
  Feng Qian, *Global Physics Summit*, March, 2026
- **Warm Start Adiabatic State Preparation**
  Feng Qian, *IPAM:Bridging the Gap Between NISQ and FTQC*, Feb, 2026
- **Adaptive Classical Shadow for Energy Estimation**
  Feng Qian, *D-Wave System Virtual Meeting*, March, 2023
- **Introduction to Quantum Simulation**
  Feng Qian, *Invited talk at the Long Table Physics Seminar, Harvard University*, October, 2023
- **Clustering Algorithm for Predicting Quantum States in a Quantum Computer**
  Feng Qian, Daniel Gauthier, *2020 OSA Frontiers in Optics and Laser Science APS/DLS(FiO+LS) Conference*, Sept.14, 2020

Honors & Awards

- OSU Physics Summer Research Scholarship. May 2019, May 2020
- College of Arts and Sciences Undergraduate Research Scholarship. Aug. 2019
- College of Arts and Sciences Dean's List. 2017–2021

Professional Activities

- Tufts Quantum Computing Seminar, Organizer

Programs and Skills

- Programming Languages: Python, Julia
- Software: OpenFermion, Qiskit, Block2, ITensor, PySCF, Cirq, Quimb

Research Interest

- Quantum Simulation, Tensor Network, Optimization

Reference

- Peter J. Love, peterjlove@gmail.com
- Alexis Ralli, alexis.ralli@qmatter.xyz